

Global Economic & Digital Development Analysis

Business Analyst Portfolio Project

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Tools Used: Google Sheets | SQL (BigQuery) | Tableau

Project Type: Global Economic & Digital Development Analysis

1. Executive Summary

This project analyses global economic and digital development trends across 10 countries between 2015 and 2024. The analysis focuses on three key indicators: GDP growth, internet usage and unemployment. Google Sheets was used for data preparation and preliminary analysis, Google BigQuery was used to perform SQL-based analysis, and Tableau Public was used to develop an interactive dashboard for visual exploration.

The analysis highlights major differences in economic performance, labour-market conditions and digital adoption across the selected countries. A particularly significant pattern is the sharp disruption in GDP growth around 2020, followed by recovery in 2021, while internet usage generally increased over the period. The project demonstrates how economic and digital indicators can be combined to identify cross-country trends, disruptions and development patterns.

2. Introduction

Economic growth, employment and digital connectivity are important indicators of a country's development and economic resilience. This project examines these dimensions using publicly available World Bank data for 10 selected countries—Australia, Brazil, China, France, Germany, India, Japan, South Africa, the United Kingdom and the United States—covering the period from 2015 to 2024.

The analysis focuses on three indicators: GDP growth (%), individuals using the Internet (% of population), and unemployment (% of total labour force). By analysing these indicators together, the project explores differences in economic performance, digital adoption and labour-market conditions across countries and over time.

The project follows an end-to-end business analytics workflow, beginning with data collection and preparation in Google Sheets, followed by SQL analysis using Google BigQuery and interactive data visualisation in Tableau Public. The final dashboard enables cross-country comparison and provides a clear visual representation of important trends across the 2015–2024 period.

3. Business Problem

Countries experience different levels of economic growth, unemployment and digital adoption, making it difficult to understand development patterns by examining a single indicator in isolation. Economic disruptions can also affect countries differently, while access to digital technologies continues to expand at varying rates.

The business problem addressed in this project is how economic performance, labour-market conditions and digital development differ across selected countries and how these indicators have changed between 2015 and 2024. Bringing the three indicators into a single analytical view enables clearer comparison of country-level trends and identification of significant changes over time.

From a business analytics perspective, the challenge is to transform multiple economic datasets into structured, comparable and visually understandable insights that can support evidence-based analysis and decision-making.

4. Project Aim and Objectives

4.1 Project Aim

The aim of this project is to analyse and compare economic and digital development trends across 10 selected countries between 2015 and 2024 using World Bank indicators. The project aims to identify meaningful patterns in GDP growth, internet usage and unemployment and communicate these findings through SQL analysis and an interactive Tableau dashboard.

4.2 Project Objectives

- Analyse GDP growth trends across the selected countries from 2015 to 2024.
- Compare internet usage levels and identify patterns in digital adoption.
- Examine unemployment trends and differences between countries.

- Identify significant changes and unusual patterns, particularly around major economic disruptions.
- Use Google BigQuery and SQL to query and analyse the prepared dataset.
- Develop an interactive Tableau dashboard to communicate the results visually.
- Generate practical, data-driven observations from the combined economic and digital indicators.

5. Dataset Overview

The project uses publicly available development indicator data from the World Bank. The dataset was prepared to provide a consistent country-year structure covering 10 selected countries between 2015 and 2024. Each record represents a country and year combination, enabling trends to be compared consistently across the analysis period.

Three indicators were selected to represent economic performance, labour-market conditions and digital development:

GDP Growth (%) – measures the annual percentage growth rate of a country's economy and is used to assess changes in economic performance over time.

Internet Usage (%) – represents the percentage of the population using the internet and is used as an indicator of digital adoption and connectivity.

Unemployment (%) – represents the percentage of the labour force that is unemployed and is used to compare labour-market conditions across countries.

The final analytical dataset contained **100 country-year observations**, representing **10 countries across 10 years (2015–2024)**.

Dataset Summary

Attribute	Details
Data Source	World Bank Open Data
Analysis Period	2015-2024
Countries	10
Country-Year Records	100

Key Indicators	GDP Growth%, Internet Usage%, Unemployment %
Granularity	One record per country per year
Final Dashboard Tool	Tableau Public

6. Data Preparation & Cleaning

6.1 Data Preparation in Google Sheets

The World Bank indicator datasets were initially prepared and reviewed in Google Sheets before being transferred into the analytical environment. The objective was to create a consistent dataset structure that could be reliably queried in SQL and visualised in Tableau.

The three indicators—GDP Growth (%), Internet Usage (%) and Unemployment (%)—were organised by country and year. The analysis was restricted to the selected 10 countries and the 2015–2024 period to maintain a consistent scope across all indicators.

6.2 Data Quality Checks

Data quality checks were performed to verify the structure and completeness of the analytical dataset. This included checking country names, year values, indicator columns and the number of observations. The prepared dataset was validated to contain 10 countries and 100 country-year records covering 2015–2024.

The data was also reviewed for formatting and consistency before analysis. Year values were standardised for time-series analysis, percentage-based indicators were maintained in a consistent numerical format, and the final structure was prepared for import into Google BigQuery.

6.3 Final Analytical Structure

The cleaned data was consolidated into a country-year analytical structure containing the fields Country Name, Country Code, Year, GDP Growth Pct, Internet Usage Pct and Unemployment Pct. This structure enabled the same dataset to support SQL queries, cross-country comparisons and Tableau visualisations without requiring separate analytical tables for each chart.

7. SQL Analysis in Google BigQuery

7.1 SQL Analysis Approach

Following data preparation, the analytical dataset was imported into Google BigQuery to perform SQL-based exploration and validation. SQL was used to confirm the scope of the dataset, calculate summary statistics, compare country-level performance and identify patterns across GDP growth, internet usage and unemployment.

Initial validation queries confirmed that the dataset contained 10 countries and covered the period from 2015 to 2024. Aggregation, filtering, grouping and ordering techniques were then used to transform the raw country-year records into analytical outputs suitable for interpretation and visualisation.

7.2 Key SQL Analysis Performed

- Validated the total number of countries and the earliest and latest years represented in the dataset.
- Calculated average GDP growth to compare economic performance across countries.
- Analysed average internet usage to compare levels of digital adoption.
- Compared unemployment rates across countries and years.
- Identified higher- and lower-performing countries for selected indicators.
- Examined annual trends to identify significant changes in economic performance.
- Prepared aggregated outputs that could be compared with the Tableau visual analysis.

7.3 SQL Skills Demonstrated

The BigQuery analysis demonstrated practical use of SQL techniques including `SELECT`, `COUNT`, `COUNT(DISTINCT)`, `MIN`, `MAX`, `AVG`, `GROUP BY`, `ORDER BY` and filtering conditions. These techniques were used to answer defined analytical questions rather than querying the dataset without a business objective.

8. Tableau Dashboard Development

8.1 Dashboard Design

Tableau Public was used to transform the analysed World Bank data into an interactive visual dashboard. The dashboard was designed to provide a clear comparison of economic, labour-market and digital-development trends across the 10 selected countries during 2015–2024.

A consistent country-based colour structure was used across the time-series visualisations, allowing users to compare how individual countries performed over the analysis period. The final dashboard brings the three core indicators into a single analytical view.

8.2 Dashboard Visualisations

- **GDP Growth Trends by Country (2015–2024):** compares annual GDP growth across the selected countries and highlights periods of economic contraction, volatility and recovery.
- **Unemployment Trends by Country (2015–2024):** compares changes in unemployment and highlights differences in labour-market performance between countries.
- **Internet Usage Trends by Country (2015–2024):** visualises changes in internet adoption and enables comparison of digital-development patterns across countries.

8.3 Interactive Dashboard

Tableau Public:

<https://public.tableau.com/app/profile/anshi.rana6970/viz/GlobalEconomicDigitalDevelopmentAnalysis/Dashboard1>

9. Key Findings & Comparative Analysis

9.1 GDP Growth Trends by Country (2015–2024)

The GDP growth chart demonstrates that economic performance was considerably more volatile than either internet usage or unemployment across the analysis period. Prior to 2020, growth rates varied between countries but generally remained within a comparatively narrower range. The most significant common disruption occurs around 2020, when several countries experienced sharp contractions, followed by pronounced rebounds during the subsequent recovery period.

The comparison also demonstrates that countries did not respond uniformly to the same global economic disruption. Some economies experienced deeper contractions and stronger subsequent rebounds, while others displayed comparatively moderate movements. This divergence highlights differences in economic structure, exposure to external shocks and recovery dynamics across the countries analysed.

After the initial recovery, growth rates generally became less extreme, although differences between countries remained visible. The chart therefore demonstrates why a country's economic performance should be evaluated across multiple years rather than using one exceptionally strong or weak annual growth figure.

Analytical interpretation: GDP growth is the most cyclical and volatile of the three indicators in the dashboard. The comparison suggests that short-term economic shocks can create substantial divergence between countries even when they are exposed to the same broader global conditions.

9.2 Internet Usage Trends by Country (2015–2024)

In contrast to the volatility observed in GDP growth, internet usage shows a much clearer and more sustained upward trajectory across the selected countries. This indicates that digital adoption represents a longer-term structural development trend rather than responding as dramatically to short-term economic fluctuations.

The chart also reveals differences in countries' starting positions. Countries that already had relatively high internet penetration at the beginning of the analysis period experienced more gradual increases as they moved closer to widespread adoption. In comparison, countries beginning from lower levels generally had greater scope for growth, resulting in more visible increases over the period.

Consequently, the gap in digital adoption between countries should not be assessed purely by the rate of increase. A country experiencing slower percentage-point growth may already have had comparatively high internet penetration, whereas faster growth may reflect a lower initial base.

Analytical interpretation: Digital development appears considerably more stable than GDP growth. The overall upward movement across countries suggests continued expansion of digital connectivity despite differences in economic performance.

9.3 Unemployment Trends by Country (2015–2024)

The unemployment chart shows greater cross-country variation than the internet-usage chart, with countries displaying different baseline unemployment levels and different degrees of fluctuation over time. Some countries maintained comparatively stable labour-market conditions, while others experienced persistently higher unemployment or more pronounced changes during the analysis period.

Changes around 2020 are particularly important when considered alongside the GDP chart. Economic contraction was accompanied by labour-market disruption in several countries, although unemployment did not respond with the same magnitude or timing everywhere. This demonstrates that a fall in GDP does not automatically translate into an identical unemployment response across economies.

The subsequent movements also indicate different labour-market recovery patterns. Countries differed in how quickly unemployment stabilised following the period of disruption, suggesting differences in economic structure, employment conditions and resilience.

Analytical interpretation: Unemployment provides a different perspective on economic performance from GDP growth. GDP may recover relatively quickly after a contraction while labour-market effects can follow a different trajectory, making both indicators necessary when evaluating economic conditions.

9.4 Comparative Analysis Across the Three Indicators

Examining all three charts together produces a more complete picture of development than analysing any indicator independently. GDP growth demonstrates substantial short-term volatility, unemployment reflects differing labour-market responses, while internet usage exhibits a comparatively consistent long-term upward trajectory.

The most important contrast is between economic volatility and digital-development stability. GDP growth changed sharply during the period of major economic disruption, whereas internet adoption continued along a broader upward path. This suggests that digital transformation can represent a structural trend that persists despite short-term changes in economic growth.

A second important comparison concerns GDP growth and unemployment. Although the indicators are economically related, the dashboard shows that their movements are not perfectly synchronised across countries. Differences in the magnitude and timing of changes demonstrate why GDP growth alone is insufficient for assessing the wider economic experience of a country.

Finally, the analysis highlights significant cross-country heterogeneity. Countries entered the period with different levels of digital connectivity and unemployment and experienced different growth trajectories. Therefore, comparisons should consider both a country's starting position and its direction of change rather than simply ranking countries using a single year's value.

9.5 Overall Analyst Insight

The combined evidence indicates that economic development is multidimensional. A country can experience volatile GDP growth while simultaneously achieving continued improvements in digital connectivity, and countries experiencing similar economic shocks can display substantially different labour-market outcomes. For decision-makers, this means that GDP growth, employment conditions and digital adoption should be assessed together when evaluating market development, economic resilience or potential investment environments.

10. Recommendations

10.1 Use Multiple Indicators for Market Assessment

Organisations assessing international markets should avoid relying on GDP growth alone when evaluating country attractiveness. The analysis demonstrates that economic growth, unemployment and digital adoption can follow different trajectories. A multi-indicator framework provides a more balanced assessment of economic momentum, labour-market conditions and digital readiness.

10.2 Distinguish Short-Term Volatility from Long-Term Trends

Sharp changes in annual GDP growth should be interpreted within a longer time horizon. The disruption and subsequent recovery visible around 2020–2021 demonstrate how individual years can produce unusually high or low growth rates. Decision-makers should therefore combine annual performance with multi-year trends before making strategic comparisons between countries.

10.3 Consider Digital Readiness in Expansion Decisions

Internet usage should be considered alongside traditional economic indicators when evaluating opportunities for digitally delivered products and services. Markets with increasing internet penetration may present expanding opportunities for e-commerce, digital financial services, online education, technology platforms and other digitally enabled business models.

10.4 Evaluate Labour-Market Conditions Separately from GDP Growth

The comparison between GDP growth and unemployment indicates that labour markets do not necessarily respond to economic changes at the same speed or magnitude. Businesses considering international expansion, workforce planning or investment should therefore assess employment conditions independently rather than assuming that stronger GDP growth automatically corresponds with stronger labour-market performance.

10.5 Use Country-Specific Analysis Rather Than a Single Global Assumption

The variation observed across the 10 countries demonstrates that economies respond differently to broader global developments. Strategic decisions should therefore incorporate country-level trends, starting conditions and recovery patterns instead of applying the same assumptions across all markets.

10.6 Develop the Dashboard as a Market Intelligence Tool

The dashboard could be updated as new World Bank data becomes available and expanded with indicators such as inflation, GDP per capita and foreign direct investment. This would enable organisations to monitor changing market conditions, compare country attractiveness and identify potential opportunities or risks when supporting international expansion and investment decisions.

Overall, the analysis supports a multidimensional approach to international market assessment. Economic growth provides insight into short-term performance, unemployment adds a labour-market perspective, and internet usage provides an indication of digital maturity. Combining these measures produces a more informative basis for strategic comparison than relying on any individual indicator.

11. Skills Demonstrated

- **Data Cleaning & Preparation:** Structured, validated and prepared World Bank indicator data in Google Sheets for analysis.
- **SQL & BigQuery:** Used SQL aggregation, filtering, grouping and ordering techniques to analyse country-level economic and digital indicators.
- **Data Visualisation:** Developed a Tableau Public dashboard comparing GDP growth, unemployment and internet usage trends across countries.
- **Trend & Comparative Analysis:** Identified differences in economic volatility, labour-market performance and digital adoption across countries and over time.
- **Business Analysis:** Translated analytical findings into business-focused insights and recommendations for market assessment and decision-making.
- **Data Storytelling:** Combined multiple indicators into a clear analytical narrative covering economic performance, labour markets and digital development.

12. Conclusion

This project demonstrates an end-to-end business analytics workflow using World Bank data to examine economic and digital development across 10 countries from 2015 to 2024. The analysis identified substantial variation in GDP growth and unemployment alongside a broader upward trend in internet adoption.

By combining Google Sheets, SQL in Google BigQuery and Tableau, the project transformed raw indicator data into structured analysis, comparative insights and an interactive dashboard. The findings demonstrate the importance of evaluating multiple indicators together when assessing economic conditions, digital maturity and potential market opportunities.

13. Project Links

Interactive Tableau Dashboard:

<https://public.tableau.com/app/profile/anshi.rana6970/viz/GlobalEconomicDigitalDevelopmentAnalysis/Dashboard1>

World Bank Data- GDP Growth (annual %):

<https://data.worldbank.org/indicator/NY.GDP.MKTP.KD.ZG>

World Bank- Individuals Using the Internet (% of population):

<https://data.worldbank.org/indicator/IT.NET.USER.ZS>

World Bank- Unemployment (% of total labour force):

<https://data.worldbank.org/indicator/SL.UEM.TOTL.ZS>

